```
import pandas as pd
import numpy as np
import matplotlib as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score,mean_squared_error
# import matplotlib.pyplot as plt
```

```
df = pd.read_csv("job_salary_prediction_dataset.csv")
df.shape
```

```
(250000, 10)
```

```
df.head()
```

	job_title	experience_years	education_level	skills_count	industry	company_size	location	remote_work	certifications
0	AI Engineer	10	Bachelor	2	Healthcare	Medium	India	Hybrid	2
1	Data Analyst	5	Bachelor	17	Telecom	Small	Australia	No	0
2	Frontend Developer	18	PhD	4	Media	Medium	Singapore	No	1

```
print("************************************")
print(df["job_title"].value_counts())
print("************************************")
print("************************************")
print(df["education_level"].value_counts())
print("************************************")
print("************************************")
print(df["industry"].value_counts())
print("************************************")
print("************************************")
print(df["company_size"].value_counts())
print("************************************")
print("************************************")
print(df["location"].value_counts())
print("************************************")
print("************************************")
print(df["remote_work"].value_counts())
print("************************************")
```

```
************************************
job_title
Backend Developer            21125
Cybersecurity Analyst        20959
Product Manager              20950
AI Engineer                  20945
Data Scientist               20890
DevOps Engineer              20889
Software Engineer            20876
Data Analyst                 20722
Cloud Engineer               20686
Machine Learning Engineer    20677
Business Analyst             20648
Frontend Developer           20633
Name: count, dtype: int64
************************************
************************************
education_level
Master         50352
High School    50065
Bachelor       49950
PhD            49857
Diploma        49776
Name: count, dtype: int64
************************************
************************************
industry
Finance          25393
Consulting       25258
Media            25034
Manufacturing    25024
```

```
Technology       24903
Government       24901
Healthcare       24898
Education        24889
Telecom          24859
Retail           24841
Name: count, dtype: int64
************************************
************************************
company_size
Large         50254
Small         50235
Medium        50027
Enterprise    49875
Startup       49609
Name: count, dtype: int64
************************************
************************************
location
Australia      25258
Canada         25165
Sweden         25100
Remote         25065
Singapore      25035
USA            24931
UK             24927
India          24895
```

```
dummy =pd.get_dummies(
    df[["job_title","education_level", "industry","company_size","location","remote_work"]],
    drop_first=True,
    dtype=int
)
dummy.head()
```

	job_title_Backend Developer	job_title_Business Analyst	job_title_Cloud Engineer	job_title_Cybersecurity Analyst	job_title_Data Analyst	job_title_Data Scientist	job_title_De Engi
0	0	0	0	0	0	0	
1	0	0	0	0	1	0	
2	0	0	0	0	0	0	
3	0	1	0	0	0	0	
4	0	0	0	0	0	0	

5 rows × 39 columns

```
ndf = pd.concat([df,dummy], axis='columns')
ndf.head()
```

	job_title	experience_years	education_level	skills_count	industry	company_size	location	remote_work	certifications
0	AI Engineer	10	Bachelor	2	Healthcare	Medium	India	Hybrid	2
1	Data Analyst	5	Bachelor	17	Telecom	Small	Australia	No	0
2	Frontend Developer	18	PhD	4	Media	Medium	Singapore	No	1
3	Business Analyst	19	PhD	13	Retail	Medium	Canada	Yes	0
4	Product Manager	15	Bachelor	7	Manufacturing	Large	Sweden	Yes	0

5 rows × 49 columns

```
ndf=ndf.drop(["job_title","education_level", "industry","company_size","location","remote_work"], axis='columns')
ndf.head()
```

	experience_years	skills_count	certifications	salary	job_title_Backend Developer	job_title_Business Analyst	job_title_Cloud Engineer	job_title_Cybe
0	10	2	2	109413	0	0	0	
1	5	17	0	93764	0	0	0	
2	18	4	1	148123	0	0	0	
3	19	13	0	189123	0	1	0	
4	15	7	0	165069	0	0	0	

5 rows × 43 columns

```
X= ndf.drop("salary", axis = 'columns')
X.head()
```

	experience_years	skills_count	certifications	job_title_Backend Developer	job_title_Business Analyst	job_title_Cloud Engineer	job_title_Cybersecuri Analy
0	10	2	2	0	0	0	
1	5	17	0	0	0	0	
2	18	4	1	0	0	0	
3	19	13	0	0	1	0	
4	15	7	0	0	0	0	

5 rows × 42 columns

```
y = ndf.salary
y.head()
```

```
0    109413
1     93764
2    148123
3    189123
4    165069
Name: salary, dtype: int64
```

```
X_train, X_test, y_train, y_test = train_test_split(X,y,test_size=.2)
model = LinearRegression()
model.fit(X_train,y_train)
```

```
▾ LinearRegression
LinearRegression()
```

```
model.score(X_test,y_test)
```

```
0.9630115687180292
```

```
y_predicted=model.predict(X_test)
y_predicted
```

```
array([115564.45132675, 129219.70335349, 177916.079122  , ...,
       166128.57418939, 129394.32579481, 181111.13242835])
```

```
r2 = r2_score(y_test, y_predicted)
r2
```

```
0.9630115687180292
```

```
mse = mean_squared_error(y_test, y_predicted)
mse
```

```
51246001.71264289
```

```
Start coding or generate with AI.
```